MATHS MANTHAN
• TUITION CLASSES •
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Subject: Mathematics | **WEEKLY CHAPTER TEST – 01 (LINEAR EQUATIONS)** | **Time:** 45 Minutes

Class: Class X (CBSE) | **Chapter 3: Pair of Linear Equations in Two Variables** | **MM: 20**

Student Name: _____ **Roll No.:** _____ **Date:** _____

GENERAL INSTRUCTIONS:

1. All questions are compulsory. Internal choice is provided in Question 9(iv).
2. **Section A** comprises 3 Multiple Choice Questions (MCQs) of 1 mark each.
3. **Section B** comprises 1 Assertion-Reason question of 1 mark.
4. **Section C** comprises 2 Very Short Answer (VSAQ) questions of 2 marks each.
5. **Section D** comprises 1 Short Answer (3M) and 1 Long Answer / HOTS (5M) question.
6. **Section E** comprises 1 Applied Case Study with 4 sub-parts of 1 mark each.

SECTION A: Multiple Choice Questions (3 × 1 = 3 Marks)

Q1. The pair of linear equations $x + 2y + 5 = 0$ and $-3x - 6y + 1 = 0$ has: [1]

(A) a unique solution (B) exactly two solutions (C) infinitely many solutions (D) no solution

Q2. If a pair of linear equations in two variables is consistent, then the lines represented by them are: [1]

(A) parallel (B) always coincident (C) intersecting or coincident (D) always intersecting

Q3. The pair of equations $y = 0$ and $y = -7$ has: [1]

(A) one solution (B) two solutions (C) infinitely many solutions (D) no solution

SECTION B: Assertion-Reason Question (1 × 1 = 1 Mark)

Q4. **Assertion (A):** The system of linear equations $9x + 3y + 12 = 0$ and $18x + 6y + 24 = 0$ has infinitely many solutions.
Reason (R): A pair of linear equations $a_1x + b_1y + c_1 = 0$ and $a_2x + b_2y + c_2 = 0$ has infinitely many solutions if $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$. [1]

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(C) (A) is true but (R) is false. (D) (A) is false but (R) is true.

SECTION C: Very Short Answer Questions (2 × 2 = 4 Marks)

Q5. On comparing the ratios $\frac{a_1}{a_2}, \frac{b_1}{b_2}$ and $\frac{c_1}{c_2}$, determine whether the pair of equations $5x - 8y + 1 = 0$ and $\frac{3}{2}x - \frac{12}{5}y + \frac{1}{3} = 0$ is consistent or inconsistent. [2]

Q6. Solve the following pair of linear equations using the substitution method: [2]

$$x + y = 14 \quad \text{and} \quad x - y = 4$$

SECTION D: Short & Long Answer Questions (1 × 3 + 1 × 5 = 8 Marks)

Q7. Solve the following pair of linear equations graphically: [3]

$$2x + y = 6 \quad \text{and} \quad 2x - y + 2 = 0$$

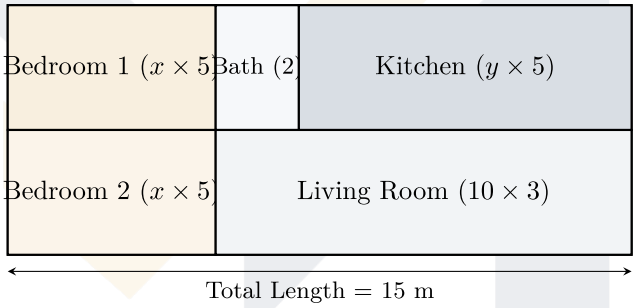
Find the coordinates of the vertices of the triangle formed by these lines and the x-axis. Also, calculate its area.

Q8. A train covered a certain distance at a uniform speed. If the train had been 10 km/h faster, it would have taken 2 hours less than the scheduled time. And, if the train were slower by 10 km/h, it would have taken 3 hours more than the scheduled time. Find the distance covered by the train. [5]

SECTION E: Applied Case Study Based Question (1 × 4 = 4 Marks)

Q9. Case Study: Architectural Layout of an Apartment (Ravish's Floor Plan)

Ravish is planning to buy a 2-BHK apartment whose layout is designed on a coordinate grid. The total area of two identical bedrooms and the kitchen together is 95 m^2. The length of each bedroom is x m and width is 5 m. The length of the kitchen is y m and width is 5 m. The front outer wall of the apartment has a total length of 15 m consisting of Bedroom 1, a 2 m wide bathroom, and the kitchen.



Based on the above geometric layout, answer the following questions:

(i) Formulate the pair of linear equations in x and y representing the outer length and the combined area of the two bedrooms and kitchen. [1]

(ii) Find the length of each bedroom (x) and the length of the kitchen (y). [1]

(iii) Calculate the total area of the living room. [1]

(iv) Find the cost of laying floor tiles in the kitchen at the rate of Rs. 50 per m^2. [1]

OR

Find the total perimeter of the combined two bedrooms. [1]

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